MCA MINI PROJECT — FIRST REVIEW

Unified Healthcare Appointment and Immunization Tracking System

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Acknowledgement

We express our sincere gratitude to the respected Review Panel for their valuable time, guidance, and insightful suggestions in evaluating our project. We would like to extend our heartfelt thanks to:

Dr. N. Ilayaraja

Ms. A. Kalyani

Dr. S. Bhama

Ms. R. Aruna

for their encouragement, constructive feedback, and expert recommendations, which will help us improve the quality of our project. We also express our sincere thanks to our project guide, the faculty members, and the Department of Computer Applications, PSG College of Technology, for their continuous support and guidance throughout this project.

Thank You.

RECAP

Project at a Glance

A web application that unifies appointment booking, automated reminders, and a patient's complete medical history across every hospital they visit — while integrating with the government-issued ID used to track a pregnancy through delivery and a child's immunization.

Since the Zeroth Review

- Finalized the 4 system roles and their access levels
- Began module-wise implementation on the MERN stack
- 5 of 8 proposed modules are now functional

PROPOSED MODULES

- Appointment booking
- Automated reminders
- Unified medical history
- Emergency data transfer
- Government ID integration
- Hospital dashboard
- Immunization Records

SYSTEM DESIGN

Roles in the System

1 Patient

Books appointments, views unified medical history, and receives automated reminders. Can link a government maternal ID for pregnancy and immunization tracking.

2 Doctor

Views assigned appointments and patient history at the point of care, and records consultation outcomes.

3 Hospital Staff

Manages appointment slots, updates patient records, and handles inter-hospital data transfer during referrals.

4 Administrator

Oversees data and activity across all participating hospitals in the system.

IMPLEMENTATION

Technology Stack

MongoDB

DATABASE

Stores patient, appointment, and record data as documents

Express.js

BACKEND FRAMEWORK

Handles REST API routing and business logic

React.js

FRONTEND

Patient and hospital-facing web interface

Node.js

RUNTIME

Powers the backend server and API layer

Together: MongoDB + Express.js + React.js + Node.js (MERN) — one JavaScript stack from database to browser.

Comments

Zeroth Review Comments

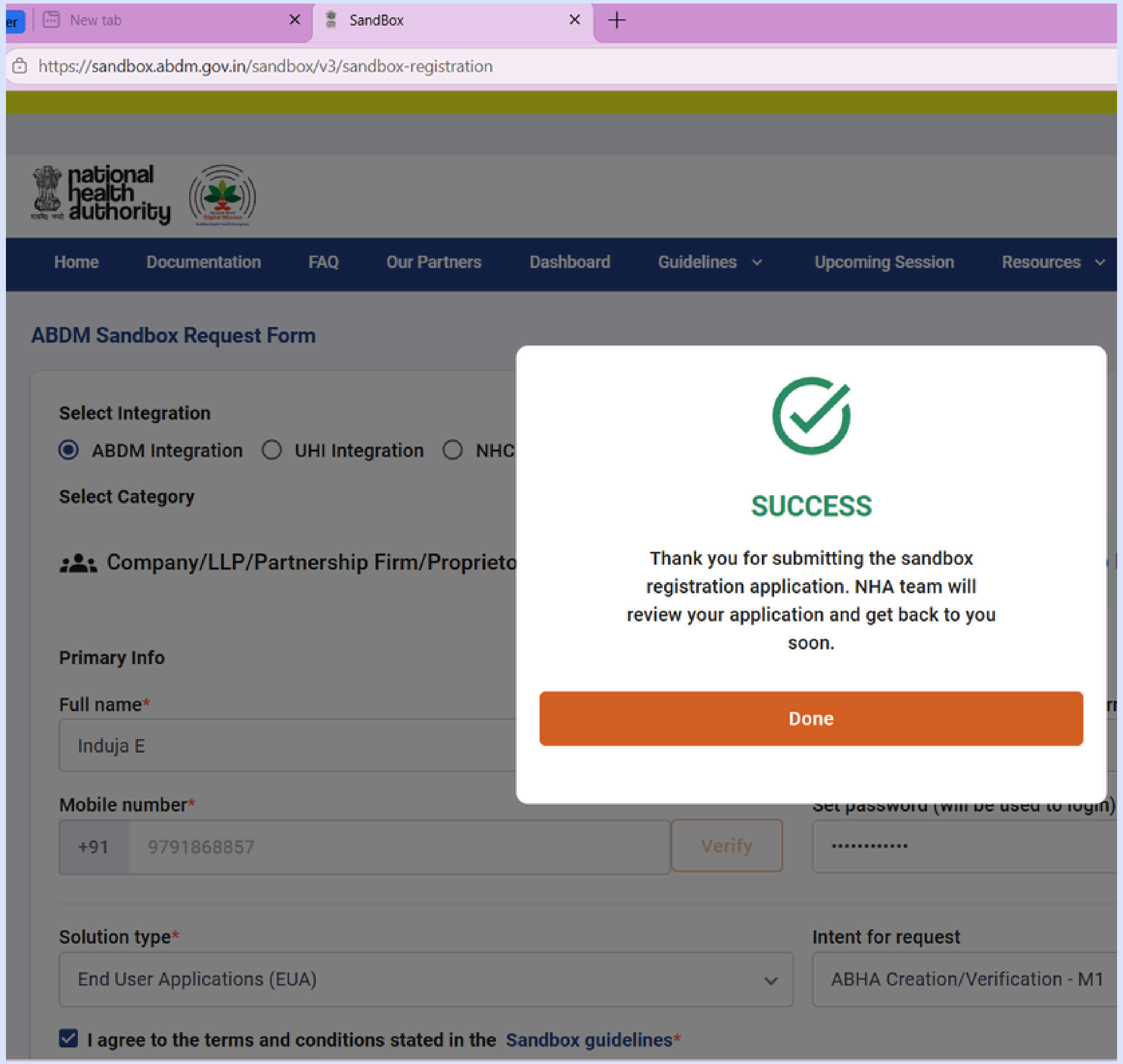
Check Feasibility of getting the government DB

ABHA ID Integration

Implemented ABHA ID registration and authentication using the ABHA Sandbox API for secure digital health identity integration.

Machine Learning Dataset Integration

Collected and integrated a Kaggle volunteer dataset for the ML module, followed by data preprocessing and model training for prediction and analysis.



PROGRESS

Modules — Progress Status

Completed

DONE	Appointment Booking
DONE	Government ID Integration
DONE	Hospital Dashboard
DONE	Immunization Records
DONE	ML Integration

Planned / Upcoming

PLANNED	Automated Reminders
PLANNED	Unified Medical History
PLANNED	Emergency Data Transfer

PROGRESS

Completed Modules

DONE

Appointment Booking

Patients search for nearby healthcare providers and book an appointment online by choosing doctor, date, time, and type.

DONE

Government ID Integration

Links a patient's antenatal, delivery, and immunization records to their existing government-issued ID.

DONE

Hospital Dashboard

Lets hospital staff view and update patient records, manage appointment slots, and log vaccination entries.

PROGRESS

Completed Modules

DONE

Immunization Records

The system maintains comprehensive vaccination tracking for patients, allowing healthcare providers to record, monitor, and retrieve immunization histories with due-date reminders for upcoming doses.

DONE

ML Integration

A Gradient Boosting machine learning model analyzes patient data — including age, medical condition, and test results — to intelligently suggest appointment types and balance doctor workloads through real-time risk scoring.

ML Integration

- **Model 1 — Appointment Type Prediction**
- **Model 2 — Doctor Risk Scoring & Workload Balancing**

What is Gradient Boosting?
A powerful ensemble learning algorithm that builds 200 decision trees sequentially, where each tree learns from the mistakes of the previous one — resulting in a highly accurate final prediction.

ML Integration

Model 1 — Appointment Type Prediction

Input: Age, Gender, Blood Type, Medical Condition, Medication, Test Results

Output: Emergency | Consultation | Follow-up | General Checkup | Vaccination

Steps:

1. Patient profile is passed as input
2. 200 trees analyze and correct each other's predictions
3. Final appointment type is suggested with confidence score

Model 2 — Doctor Risk Scoring & Workload Balancing

Input: Age, Medical Condition, Test Results

Output: Low | Moderate | High | Critical

Steps:

1. Each active patient of a doctor is risk-scored (0–100)
2. Average risk score is calculated per doctor
3. Doctor with lowest score is recommended for new appointment

Database Models — Draft Schema

Model	Key Fields
User	name, email, password, role, hospitalId, contactInfo
Appointment	patientId, doctorId, hospitalId, date, time, type, status, notes
MaternalRecord	govtMaternalId, patientId, antenatalVisits[], deliveryDetails, immunizationLinks
Slot	hospitalId, doctorId, date, time, isBooked
Vaccination	patientId, vaccineName, date, status, administeredBy

Draft schema based on modules proposed at the Zeroth Review — to be aligned with the actual implementation.

IMPLEMENTATION

API Endpoints — Completed Modules

Method	Route	Purpose
GET	/api/appointments/search	Search nearby healthcare providers
POST	/api/appointments	Book an appointment
GET	/api/maternal/:govtId	Look up a maternal record by government ID
POST	/api/maternal	Register a maternal record and link the government ID
POST	/api/hospital/slots/:id	Update an appointment slot
GET	/api/hospital/vaccinations	Log a vaccination entry

Sample route naming for the three completed modules — adjust to match your actual API.

ROADMAP

Remaining Work

1

Automated Remainders

Alerts for check-ups, follow-ups, and medication schedules.

2

Unified Medical History

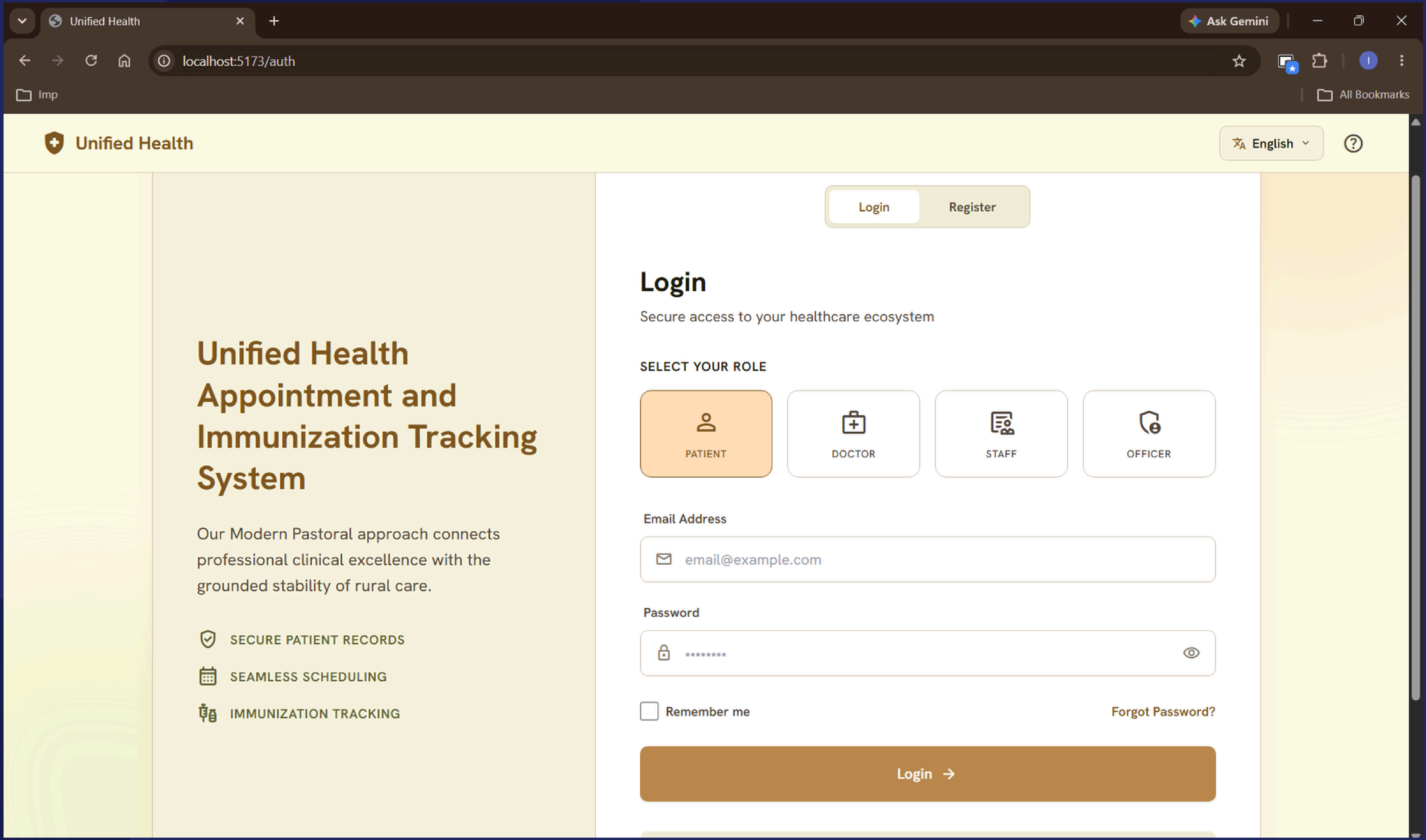
Consolidates diagnoses, treatment, and test records across hospitals.

3

Emergency Data Transfer

Shares patient history with a receiving hospital during referral.

Login Page


Unified Health
localhost:5173/auth
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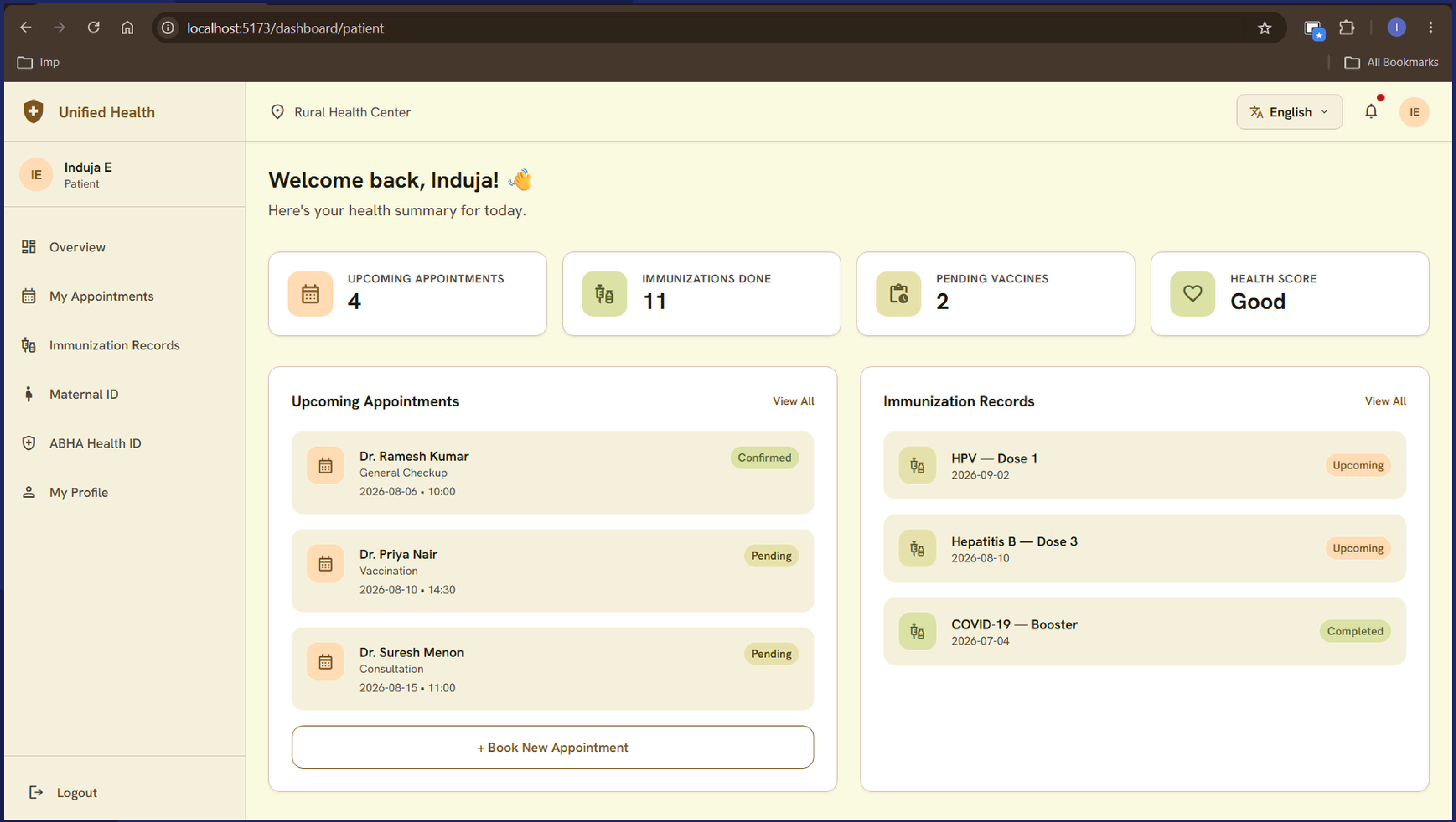
Unified Health
English
Login
Register
Login
Secure access to your healthcare ecosystem
SELECT YOUR ROLE
PATIENT
DOCTOR
STAFF
OFFICER
Email Address
email@example.com
Password
Remember me
Forgot Password?
Login
Unified Health Appointment and Immunization Tracking System
Our Modern Pastoral approach connects professional clinical excellence with the grounded stability of rural care.
SECURE PATIENT RECORDS
SEAMLESS SCHEDULING
IMMUNIZATION TRACKING

Dashboard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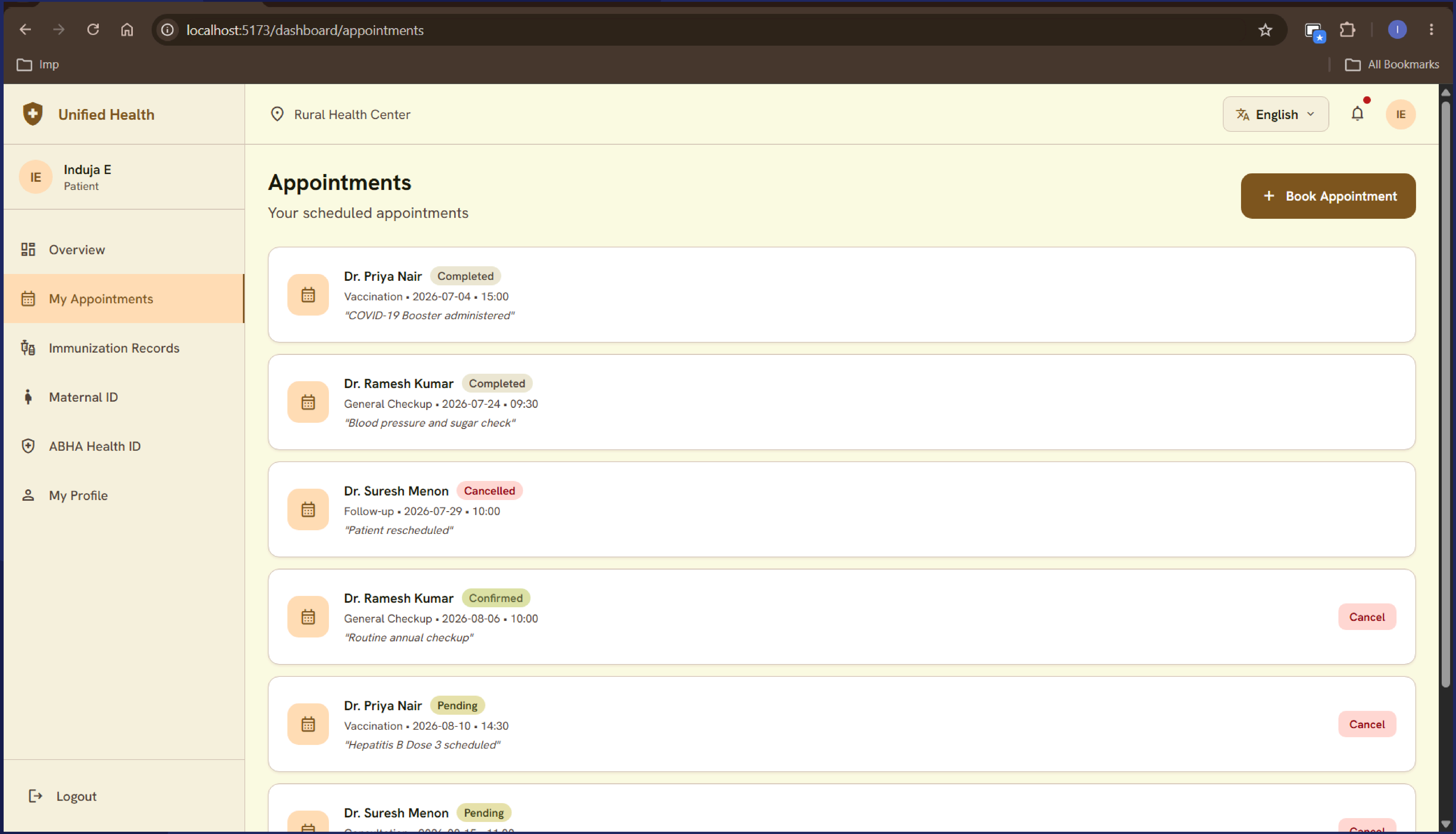
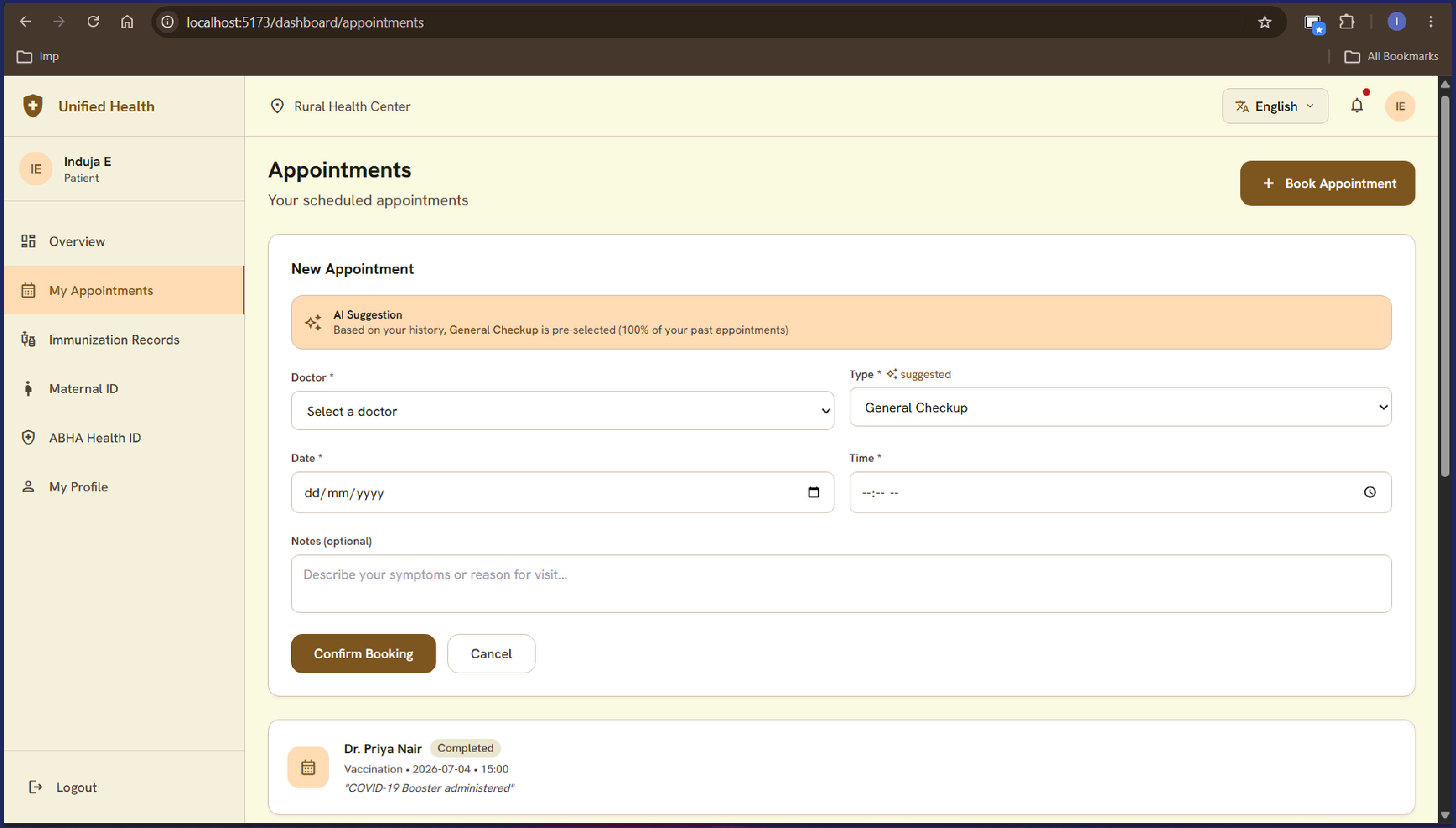
Patient
Overview
My Appointments
Immunization Records
Maternal ID
ABHA Health ID
My Profile
Logout
Welcome back, Induja! 👋
Here's your health summary for today.
UPCOMING APPOINTMENTS
4
IMMUNIZATIONS DONE
11
PENDING VACCINES
2
HEALTH SCORE
Good
Upcoming Appointments
View All
Dr. Ramesh Kumar
General Checkup
2026-08-06 • 10:00
Confirmed
Dr. Priya Nair
Vaccination
2026-08-10 • 14:30
Pending
Dr. Suresh Menon
Consultation
2026-08-15 • 11:00
Pending
+ Book New Appointment
Immunization Records
View All
HPV — Dose 1
2026-09-02
Upcoming
Hepatitis B — Dose 3
2026-08-10
Upcoming
COVID-19 — Booster
2026-07-04
Completed

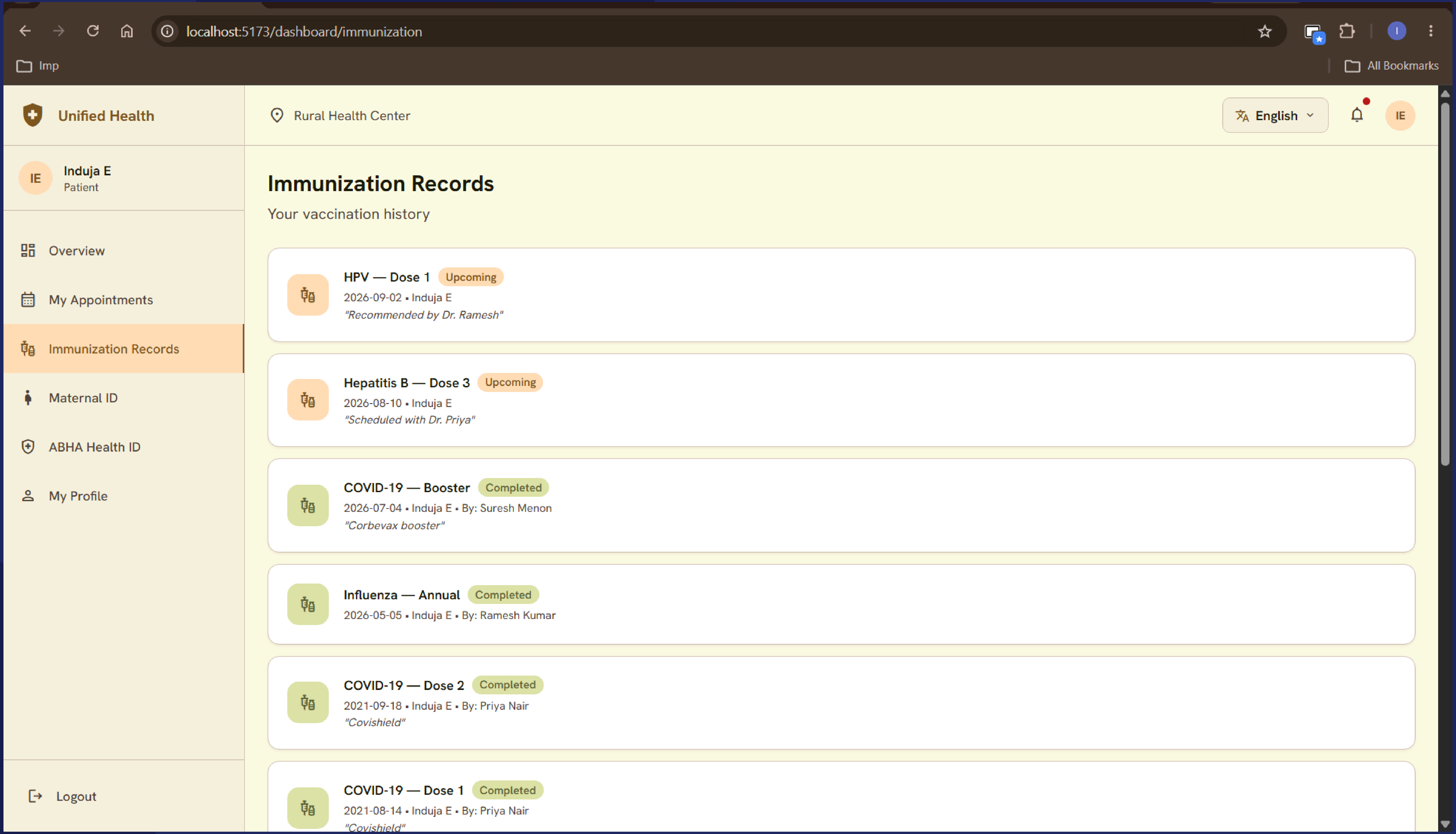
Appointments



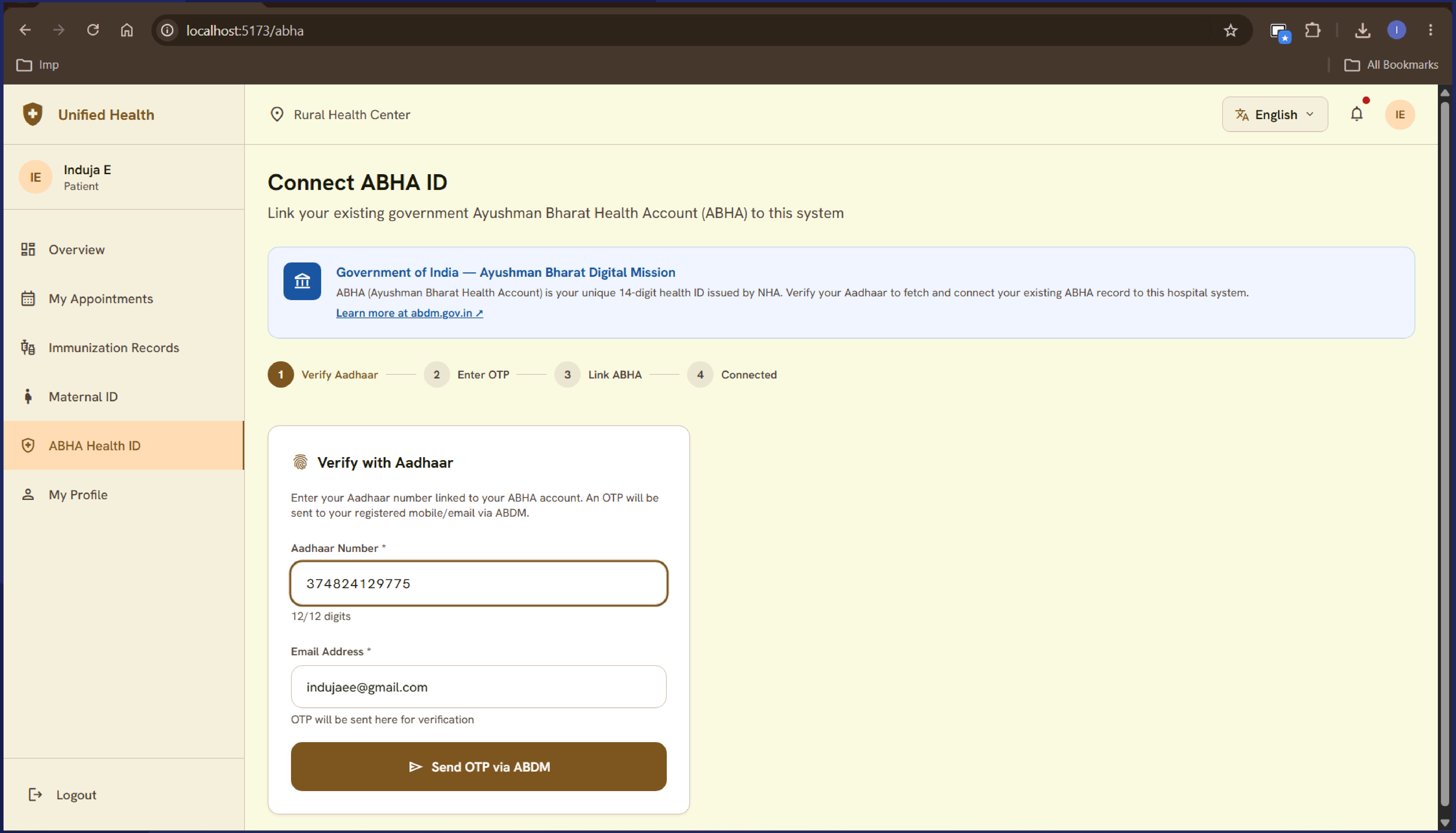
Appointment suggestions



Immunization Records



ABHA ID



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All Bookmarks
Unified Health
Rural Health Center
English
IE
Induja E
Patient
Overview
My Appointments
Immunization Records
Maternal ID
ABHA Health ID
My Profile
Logout
Connect ABHA ID
Link your existing government Ayushman Bharat Health Account (ABHA) to this system
Government of India — Ayushman Bharat Digital Mission
ABHA (Ayushman Bharat Health Account) is your unique 14-digit health ID issued by NHA. Verify your Aadhaar to fetch and connect your existing ABHA record to this hospital system.
Learn more at abdm.gov.in ↗
1 Verify Aadhaar
2 Enter OTP
3 Link ABHA
4 Connected
Verify with Aadhaar
Enter your Aadhaar number linked to your ABHA account. An OTP will be sent to your registered mobile/email via ABDM.
Aadhaar Number *
374824129775
12/12 digits
Email Address *
indujaee@gmail.com
OTP will be sent here for verification
Send OTP via ABDM

Demo Video

Unified Health | Inbox (3,562) - indujaee@gmail.com

localhost:5173/auth

Unified Health

English

Unified Health Appointment and Immunization Tracking System

Our Modern Pastoral approach connects professional clinical excellence with the grounded stability of rural care.

- SECURE PATIENT RECORDS
- SEAMLESS SCHEDULING
- IMMUNIZATION TRACKING

Login | Register

Login

Secure access to your healthcare ecosystem

SELECT YOUR ROLE

PATIENT | DOCTOR | STAFF | OFFICER

Email Address

indujaee@gmail.com

Password

••••••••

Remember me

Forgot Password?

Login →

OPTIMIZED FOR LOW-BANDWIDTH RURAL CONNECTIONS

Queries & Suggestions

Thank You